Dynamical Models

EART22001 / EART60071

1 From forces to fluid motion

Environmental fluids such as the atmosphere and ocean obey the same basic mechanics as other physical systems, but their motion is distributed continuously through space. Dynamical models translate those physical laws into equations for quantities such as velocity, pressure and fluid depth, and then solve those equations numerically on a grid.

In this chapter we build the ideas needed for the shallow-water and other fluid-dynamical models used later in the module. By the end you should be able to:

- connect pressure-gradient and Coriolis terms to horizontal acceleration;
- explain the difference between geostrophic balance and gradient-wind balance;
- interpret vorticity as a measure of local rotation in a flow;
- use the shallow-water gravity-wave speed and state the assumptions behind it;

We begin with a short review of Newtonian mechanics and then apply it to a simplified two-dimensional rotating fluid.

2 Newton's mechanics

2.1 Newton's laws (equations) of motion

Let us briefly review Newton's three laws of motion for the motion of objects with constant mass.

- ***First law.*** if the *net force* is zero then the *velocity* of the object is constant. This applies in a vector sense:
$$\sum \mathbf{F} = 0 \Leftrightarrow \frac{d\mathbf{v}}{dt} = 0 \tag{1}$$
Note that the $\sum$ symbol means to add up *all* forces acting on the object.

- ***Second law.*** Perhaps the most famous law states that the *net force* acting on an object is equal to the product of mass and acceleration:
$$\mathbf{F} = m\mathbf{a} \tag{2}$$
$$= m\frac{d\mathbf{v}}{dt} \tag{3}$$

- ***Third law.*** All forces exist in pairs: if one object A exerts a force $\mathbf{F}_A$ on a second object B, then B simultaneously exerts a force $\mathbf{F}_B$ on A and the two forces are equal size and opposite sign: $\mathbf{F}_A = -\mathbf{F}_B$.

2.2 Examples of Forces

First we consider an objects weight, W, which is an example of a force directed towards the centre of earth.

- If the object's position is in mid-air then, applying Newton's Second Law, this *downward force* gives rise to an *acceleration towards* the centre of earth.
- If the object is placed on a table its position will be fixed. Its weight still acts downward, but the table exerts an upward *normal contact force*. When these two forces are equal in magnitude the net force on the object is zero. Note that the normal force and the object's weight are *not* a Newton's-third-law pair: the partner to the table's force on the object is the object's equal and opposite force on the table.
- If the object falls through air then the object will have a *drag force*, which opposes its motion and is due to the air pushing past it. When a certain speed is reached the drag force will equal the weight force and therefore there will be *no net force* acting on the falling object. The object will still be travelling towards the ground but, in accordance with Newton's First and Second Laws, not accelerating.

Next, let us consider pressure differences (pressure gradients) giving rise to forces. Consider an air-tight room where the pressure inside the room is equal to the pressure outside the room and that there is a window, area A, on one of the walls of the room. At this point there will be *no net force* on the window; however, if we *increase* the pressure inside the room there will be a *pressure gradient* across the window. This results in a force on the window, perpendicular to the plane of the window and directed outwards:

$$F = \Delta P \times A \tag{4}$$

The key points here are that both an objects *weight* and also *pressure gradients* are associated with *forces*, which, through Newton's Second Law, can result in *accelerations*.

2.3 Uniform circular motion

Fluids tend to swirl around in ways that can be roughly approximated by circles. Here we briefly examine the requirements for *uniform circular motion*, which is motion in a circle at constant tangential speed.

First let's consider an object on a string moving in a uniform circle, with one end of the string at the centre, radius R, with angular frequency, $\omega = \frac{2\pi}{T}$, where T is the time it takes for one complete revolution. Mathematically its position versus time can be described as:

$$x = R\cos\omega t \tag{5}$$
$$y = R\sin\omega t \tag{6}$$

As said above objects in motion must obey *Newton's laws of motion*; hence, we may determine the *net force* that must act on the mass to give rise to this circular motion: we just have to find the acceleration and put the result into Newton's second law. The acceleration is the 2nd derivative of position:

$$\ddot{x} = -\omega^2 x \tag{7}$$
$$\ddot{y} = -\omega^2 y \tag{8}$$

For the object on a string, the tension provides the inward acceleration required to keep changing the direction of the velocity. If the outward radial direction is taken as positive, the radial acceleration is

$$a_r = -\omega^2 R \tag{9}$$
$$= -\frac{v_t^2}{R} \tag{10}$$

where $R = \sqrt{x^2+y^2}$. Thus the *magnitude* of the centripetal acceleration is v_t^2/R, and its direction is towards the centre of the circle. The speed can remain constant even though the velocity is changing, because velocity also contains direction.

We have also made use of the fact that, for a constant radius circle, the time it takes for one revolution is the distance, $2\pi R$, divided by the speed, v_t: $T = \frac{2\pi R}{v_t}$; hence, $\frac{v_t}{R} = \omega$.

3 Horizontal momentum equations in a rotating fluid

For a simplified horizontal, two-dimensional fluid on a rotating Earth, Newton's second law can be written in component form as

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = \rho f v - \frac{\partial P}{\partial x} \tag{11}$$
$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\rho f u - \frac{\partial P}{\partial y}. \tag{12}$$

These equations are deliberately simplified: friction, vertical motion and density variations are omitted here so that we can focus on the two dominant horizontal terms used in the practical. The terms in parentheses form the *material acceleration*: they describe how the velocity of a moving fluid parcel changes with time. Multiplication by density converts acceleration per unit mass into force per unit volume.

Here x and y are the eastward and northward coordinates, u and v are the corresponding velocity components, P is pressure, ρ is density and f is the Coriolis parameter.

3.1 Pressure-gradient force

The terms $-\partial P/\partial x$ and $-\partial P/\partial y$ are the components of the pressure-gradient force *per unit volume*. The minus sign is important: the force points from higher pressure towards lower pressure. In one dimension its magnitude can be estimated by

$$PGF_{\mathrm{vol}} \simeq \frac{|\Delta P|}{\Delta x}. \tag{13}$$

Its units are Pa m^{-1}, which are equivalent to N m^{-3}. If we divide by density, $-(1/\rho)\nabla P$, we obtain the pressure-gradient acceleration (force per unit mass), with units m s^{-2}.

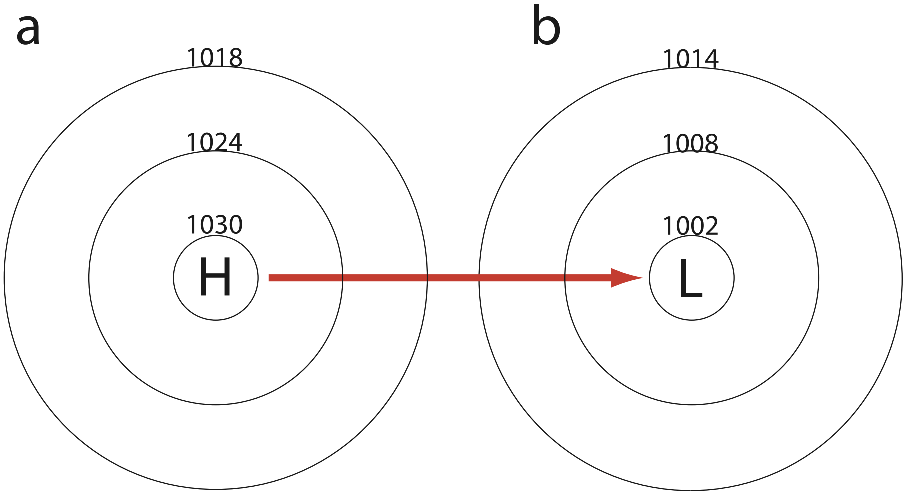


Figure 1: Schematic depicting the pressure gradient force from a high pressure system (a) to a low pressure system (b). The circles are lines of constant pressure. This is in a non-rotating system.

This is shown in Figure 1. In the absence of any other forces air will move from high pressure to low pressure in a straight line.

3.2 Coriolis effect or 'force'

The terms $\rho f v$ and $-\rho f u$ represent the Coriolis term in the rotating frame of reference attached to Earth. In an inertial frame there is no physical interaction producing a new force; rather, the Coriolis term is introduced because our coordinate system is rotating. It is therefore often called an *apparent* or *fictitious* force.

For horizontal motion on Earth the Coriolis parameter is

$$f = 2\Omega \sin\phi, \tag{14}$$

where $\Omega \simeq 7.292 \times 10^{-5}$ s^{-1} is Earth's angular rotation rate and ϕ is latitude. Thus $f = 0$ at the equator, is positive in the Northern Hemisphere and negative in the Southern Hemisphere.

The Coriolis force per unit volume in the two horizontal directions is

$$F_x = \rho f v, \tag{15}$$
$$F_y = -\rho f u. \tag{16}$$

In the Northern Hemisphere the Coriolis acceleration is to the right of the motion; in the Southern Hemisphere it is to the left. The Coriolis term changes the *direction* of a parcel's motion but, by itself, does no work on the parcel because it is perpendicular to the velocity.

This deflection is why large-scale atmospheric flow tends to turn rather than accelerate directly from high pressure to low pressure. In Figure 2, the pressure-gradient force starts the air moving across the pressure field and the Coriolis term turns that motion, producing the familiar circulation around pressure systems.

3.3 Acceleration of a fluid parcel

The left-hand sides of Equations 11 and 12 contain both a local time derivative and *advection* terms. Together they form the material derivative: the acceleration experienced by a parcel as it moves through a velocity

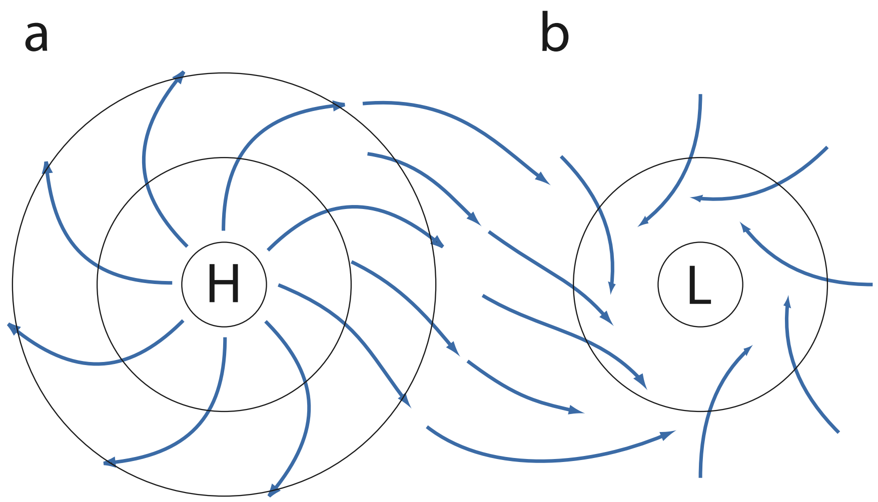


Figure 2: Schematic depicting why air flows clockwise around a high pressure system (a) and anticlockwise around a low pressure system (b). Note this diagram is for the northern hemisphere, it is the opposite in the southern hemisphere.

field. We will not derive the material derivative in detail here, but it is worth recognising why a fluid can accelerate even when the velocity at a fixed point is steady: the parcel may move into a region where the velocity is different.

4 Balanced flow

Two useful approximations illustrate how the momentum equations can be simplified. In *geostrophic balance*, the horizontal material acceleration is small and the pressure-gradient and Coriolis terms nearly cancel. In *gradient-wind balance*, the flow is steady but curved, so a centripetal acceleration remains. These are approximations, not separate laws of nature, and their usefulness depends on the spatial and temporal scales of the flow.

4.1 Geostrophic wind

$$\cancel{\frac{\partial u}{\partial t}} + \cancel{u\frac{\partial u}{\partial x}} + \cancel{v\frac{\partial u}{\partial y}} = fv - \frac{1}{\rho}\frac{\partial P}{\partial x} \tag{17}$$

$$\cancel{\frac{\partial v}{\partial t}} + \cancel{u\frac{\partial v}{\partial x}} + \cancel{v\frac{\partial v}{\partial y}} = -fu - \frac{1}{\rho}\frac{\partial P}{\partial y} \tag{18}$$

which results in:

$$u_g = -\frac{1}{\rho f}\frac{\partial P}{\partial y} \tag{19}$$

$$v_g = \frac{1}{\rho f}\frac{\partial P}{\partial x} \tag{20}$$

The resulting *geostrophic wind* is therefore parallel to the isobars rather than directed from high to low pressure. It is a good first approximation for large-scale, slowly varying flow away from the equator and away from strong frictional effects.

4.2 Gradient wind

Gradient-wind balance extends the geostrophic approximation to curved flow around low- and high-pressure systems. The speed is steady along an ideal circular path, but the direction changes, so the parcel has a centripetal acceleration of magnitude v_{gr}^2/R (Equation 10). Balancing this acceleration against the pressure-gradient and Coriolis terms produces a quadratic equation for the wind speed.

For a low pressure system:

$$\frac{v_{gr}^2}{R} = -f v_{gr} + \frac{1}{\rho}\frac{|\Delta P|}{R} \tag{21}$$

where v_{gr} is the gradient wind; R is the radius that air travels around. The last term is the pressure gradient force. For cyclonic flow around a low in the Northern Hemisphere, the inward pressure-gradient force is larger than the outward Coriolis term. The difference supplies the required inward centripetal acceleration, so the gradient wind is typically slightly slower than the corresponding geostrophic wind.

For a high pressure system:

$$\frac{v_{gr}^2}{R} = f v_{gr} - \frac{1}{\rho}\frac{|\Delta P|}{R} \tag{22}$$

For anticyclonic flow around a high in the Northern Hemisphere, the inward Coriolis term must exceed the outward pressure-gradient force, so the near-geostrophic gradient wind is faster than the geostrophic estimate. The quadratic equation also shows that sufficiently strong curvature/pressure gradients can remove the usual anticyclonic solution.

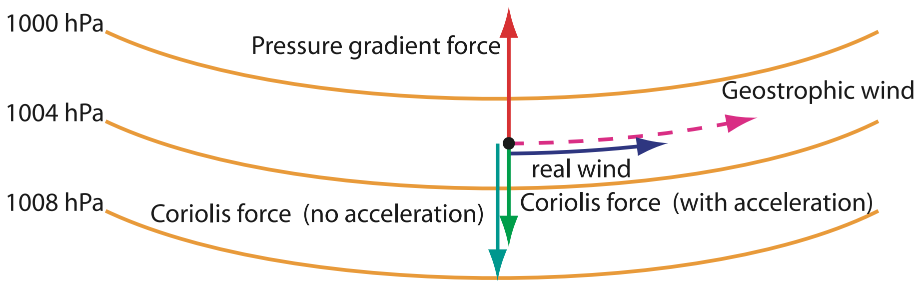


Figure 3: Schematic of forces on air flowing around a low pressure system. The pressure gradient force acts towards the low, but the Coriolis forces acts in the opposite direction. For gradient wind flow the pressure gradient must exceed the Coriolis force so that the acceleration toward the centre is large enough for circular motion.

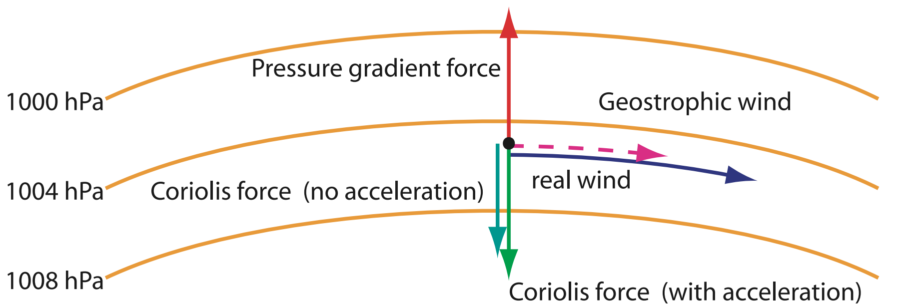


Figure 4: Schematic of winds flowing around a high pressure system. In this case the Coriolis force has to exceed the pressure gradient force; hence, winds have to be higher than geostrophic.

5 Vorticity

We define u as the *east-west* component of the wind and v as the *south-north* component. We can write the wind as a vector quantity:

$$\mathbf{v} = \begin{pmatrix} u \\ v \end{pmatrix} \tag{23}$$

where u and v can be defined as functions of position x and y for example (or something else).

Vorticity, ζ, measures the local tendency for a fluid element to rotate. The flow does not have to travel around a single centre: shear can also produce vorticity. A useful mental picture is to place a tiny paddle wheel in the flow; if it tends to rotate, the local flow has non-zero vorticity. For two-dimensional horizontal motion, the vertical component of relative vorticity is

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \tag{24}$$

6 Waves in the atmosphere

6.1 Gravity waves / Tsunamis

Gravity waves occur when gravity or buoyancy provides a restoring force after a fluid is displaced from equilibrium. In the atmosphere they can be generated, for example, by flow over mountains or by strong convection and frontal systems. Tsunamis are an important shallow-water example in the ocean.

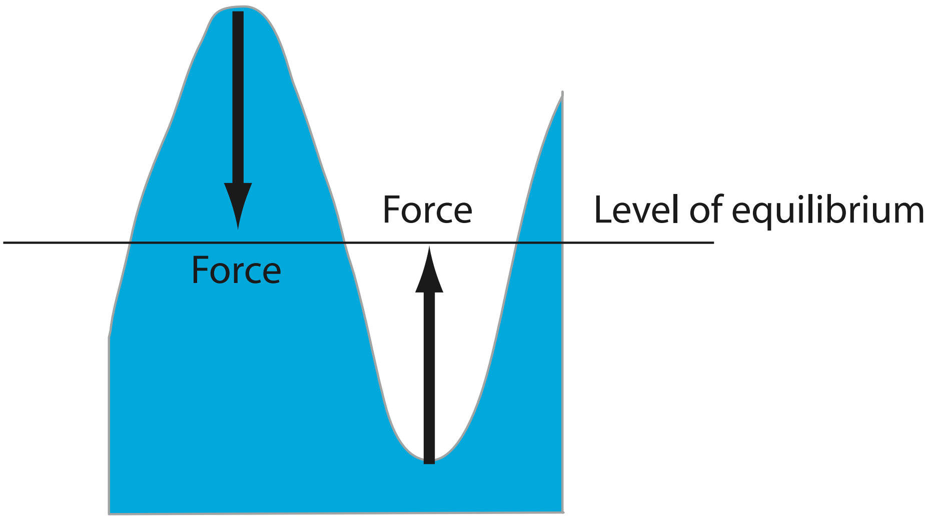


Figure 5: Schematic of a water gravity wave. In the peaks gravity provides the force to push the peak back to equilibrium. As gravity pushes the peaks down water moves into the trough and pushes the trough up.

For a long, small-amplitude shallow-water gravity wave, where the horizontal wavelength is much larger than the fluid depth, the phase speed is approximately

$$c = \sqrt{gh}. \tag{25}$$

Here c is the wave speed, $g = 9.81$ m s^{-2} is gravitational acceleration and h is the mean fluid depth. An important feature of this shallow-water limit is that the speed depends on depth but not directly on wavelength. It is interesting to see how gravity waves propagate as the depth of the fluid changes, like when a Tsunami approaches the shore. This will be investigated in the practical.

6.2 Waves associated with jets

Vorticity in a fluid tends to persist. However, vorticity does not have to mean air rotates in circles, it can also mean the air has *shear*. The *jet stream* is an example of this: in the jet the air is moving very fast west-east and the winds decrease away from the jet; thus on the north side of the jet the air has positive vorticity and on the south side it has negative vorticity. If air close to the jet moves further away from the jet then it will find itself in a region of lower shear; however, vorticity still needs to maintained. Hence, the air will rotate (anti-clockwise for positive vorticity and clockwise for negative vorticity). This regular spacing of regions of rotating air is known as *barotropic instability* and can give rise to cyclones and anticyclones. See Figure 6

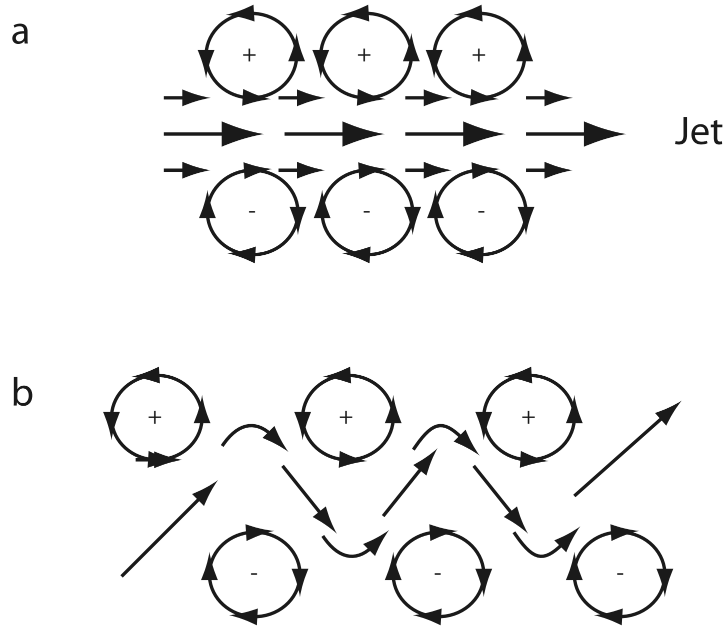


Figure 6: Schematic of barotropic instability, where cyclones and anticyclones form either side of a jet. (a) Initial development; (b) later stages of wave.

6.3 Rossby waves

When fluid flows up a mountain any vortices will be *compressed* and spread out side ways. This slows the rotation of the vortex and cause the *relative vorticity* to be negative. As the fluid flows off the mountain any vortices will be *stretched* in the vertical, which leads to an increase in the spin and positive vorticity, thus setting up a wave known as a *Rossby wave* (one that forms because of the conservation of vorticity). The effect is similar to when a ballet dancer starts to spin low down with their arms held out, but increases the rate of spin by standing tall and bringing their arms in towards their centre of mass. Rossby waves are a vital component of our weather.

An important point is air that is far north has a lot of absolute vorticity (because of the spin of earth on its axis). If it moves south, where the atmosphere is deeper, or become stretched in another way, it will conserve this vorticity and start to spin anticlockwise; thus moving north. This would not happen on a rotating cylindrical planet, where the planetary vorticity does not depend on distance from the equator.

6.4 Equatorial waves

Equatorial waves form because the Coriolis parameter goes through zero at the equator. One type is an instability that forms either side of a jet, similar to barotropic instability. However, because the Coriolis parameter changes sign, once the air crosses the equator it starts to slow down and rotate in the opposite direction. This does not happen with barotropic instability. Hence, these waves have a different name: *mixed Rossby-gravity waves*

In the practical you will test this using an *equatorial beta-plane*, which is a set-up that models the region close to the equator using a Coriolis parameter that is positive above the equator and negative below the equator.

Another kind of equatorial wave is the *equatorial Kelvin wave.* These waves form because the Coriolis forces acts to the right of air's motion in the northern hemisphere and to the left of the air's motion in the southern hemisphere. Thus if air / fluid is moving east there will be convergence towards the equator. Kelvin waves are only possible for air that moves towards the east on planets with anticlockwise rotation.

7 Questions to go through in class

7.1 Equations of motion in the atmosphere

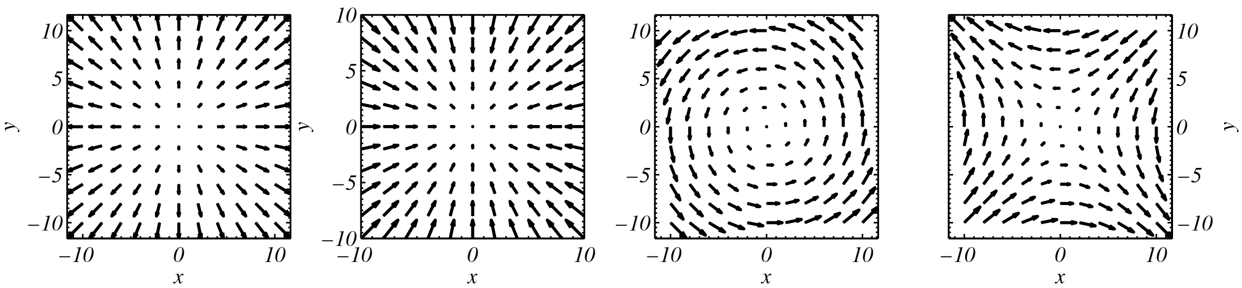
1. In the atmosphere the pressure decreases going east to west from 1018 hPa to 1008 hPa over a distance of 1000 km. Using Equation 13 what will be the pressure gradient force per unit volume? Which direction does the force act?
2. Calculate the Coriolis parameter, f, at the following latitudes: $\phi = [50°, 0°, -50°]$.
3. Calculate the Coriolis force at $\phi = 50°$ for $u = 10$ m s^{-1}. Calculate the force for $\phi = -50°$ and $u = 10$ m s^{-1}. (assume $v = 0$ and $\rho = 1$ kg m^{-3}). (use Equation 16)
4. Calculate the Coriolis force at $\phi = 50°$ for $v = 10$ m s^{-1}. Calculate the force for $\phi = -50°$ and $v = 10$ m s^{-1} (use Equations 15).

7.2 Balanced flow

1. **Geostrophic wind.** The change in pressure at $\phi = 50°$ north, where the Coriolis parameter is 1×10^{-4} s^{-1}, is $\Delta P = 500$ Pa over a distance $\Delta y = 1000 \times 10^3$ m. Assuming the density of air is $\rho = 1$ kg m^{-3}, calculate the geostrophic wind (use Equation 19 and approximate $-\frac{\partial P}{\partial y}$ by dividing ΔP by Δy).
2. **Gradient wind 1.** Use the same conditions as above but assume that the air is rotating in a circle, radius Δx, around a low pressure centre. Use Equation 21 to estimate the gradient wind.
3. **Gradient wind 2.** Use the same conditions as above but assume that the air is rotating in a circle, radius Δx, around a high pressure system. Use Equation 22 to estimate the gradient wind.

7.3 Vorticity

Calculate the vorticity of the following vector fields using Equations 23 and 24:



1. $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$. (see above, 1st plot on left).
2. $\mathbf{v} = \begin{pmatrix} -x \\ -y \end{pmatrix}$ (see above, 2nd plot from left).
3. $\mathbf{v} = \begin{pmatrix} -y \\ x \end{pmatrix}$ (see above, 3rd plot from left).
4. $\mathbf{v} = \begin{pmatrix} -y \\ -x \end{pmatrix}$ (see above, last plot on right).

7.4 Waves

All of these questions require the use of Equation 25.

1. A wave in the ocean has depth of 1 km. Calculate its phase speed.
2. The same wave approaches the coast-line where the depth is 2 m. What is its phase speed now?
3. Assuming the depth of the atmosphere is 10 km, how fast do gravity waves propagate?

8 Answers to questions

1. The pressure difference is 10 hPa = 1000 Pa, so the pressure-gradient-force magnitude is

$$\frac{1000}{1000 \times 10^3} = 1.0 \times 10^{-3}\ \mathrm{N\,m^{-3}}.$$

   Pressure is lower to the west (1008 hPa) than to the east (1018 hPa), so the force acts **westward**, from high pressure towards low pressure.
2. Using $f = 2\Omega \sin\phi$ with $\Omega = 7.292 \times 10^{-5}$ s^{-1} (watch the degree/radian setting), we obtain $f = 1.117 \times 10^{-4}$, 0, and -1.117×10^{-4} s^{-1} at 50°, 0°, and −50°, respectively.
3. We calculated the Coriolis parameters above for $\phi = 50°$ and $\phi = -50°$; hence multiply these by the density (1 kg m^{-3}) and the velocity to get $F_y = -1 \times 1.117 \times 10^{-4} \times 10 = -1.117 \times 10^{-3}$ N m^{-3} and $F_y = -1 \times -1.117 \times 10^{-4} \times 10$ N m^{-3} $= 1.117 \times 10^{-3}$. Hence, in both cases the force is towards the equator.
4. We calculated the Coriolis parameters above for $\phi = 50°$ and $\phi = -50°$; hence multiply these by the density (1 kg m^{-3}) and the velocity to get $F_x = 1 \times 1.117 \times 10^{-4} \times 10 = 1.117 \times 10^{-3}$ N m^{-3} and $F_x = 1 \times -1.117 \times 10^{-4} \times 10$ N m^{-3} $= -1.117 \times 10^{-3}$. Hence, in the northern hemisphere the force is towards the right of its motion and in the southern it is towards the left of its motion.

8.1 Balanced flow

1. **Geostrophic wind.** The magnitude is the pressure-gradient acceleration divided by the Coriolis parameter:

$$\begin{aligned} |u_g| &= \frac{1}{1 \times 10^{-4}} \frac{500}{1000 \times 10^3} \\ &= 5\ \mathrm{m\ s^{-1}}. \end{aligned}$$

   The question specifies the magnitude of the pressure change but not which side has the higher pressure, so this determines the *speed*; the wind direction follows from the sign of $-\partial P/\partial y$ in Equation 19.
2. **Gradient wind 1.** You get:

$$\frac{v_{gr}^2}{1000 \times 10^3} = -1 \times 10^{-4} v_{gr} + \frac{500}{1000 \times 10^3}$$

   Solving as a quadratic $ax^2 + bx + c = 0$ with $a = 1 \times 10^{-6}$, $b = 1 \times 10^{-4}$ and $c = -5 \times 10^{-4}$ gives that $x = -104.77$ or $x = 4.77$. The solution is the positive value, so $v_{gr} = 4.77$ m s^{-1}.
3. **Gradient wind 2.** You get:

$$\frac{v_{gr}^2}{1000 \times 10^3} = 1 \times 10^{-4} v_{gr} - \frac{500}{1000 \times 10^3}$$

   Solving as a quadratic $ax^2 + bx + c = 0$ with $a = 1 \times 10^{-6}$, $b = -1 \times 10^{-4}$ and $c = 5 \times 10^{-4}$ gives $x = 94.72$ or $x = 5.28$. The quadratic has two mathematical roots; the smaller, near-geostrophic branch is the relevant solution for this exercise, so $v_{gr} = 5.28$ m s^{-1}.

8.2 Vorticity

Answers are:

1. $\zeta = \frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} = 0$
2. $\zeta = \frac{\partial -y}{\partial x} - \frac{\partial -x}{\partial y} = 0$
3. $\zeta = \frac{\partial x}{\partial x} - \frac{\partial -y}{\partial y} = 1 - (-1) = 2$
4. $\zeta = \frac{\partial -x}{\partial x} - \frac{\partial -y}{\partial y} = -1 - (-1) = 0$

8.3 Waves

1. $\sqrt{9.81 \times 1000} = 99.0$ m s^{-1}
2. $\sqrt{9.81 \times 2} = 4.43$ m s^{-1}
3. $\sqrt{9.81 \times 10000} = 313$ m s^{-1}. This is of the same order as the speed of sound.